Title of the Project:
E-Commerce Platform for Online Buying and Selling

Name of Supervisor:
Faisal Hussain

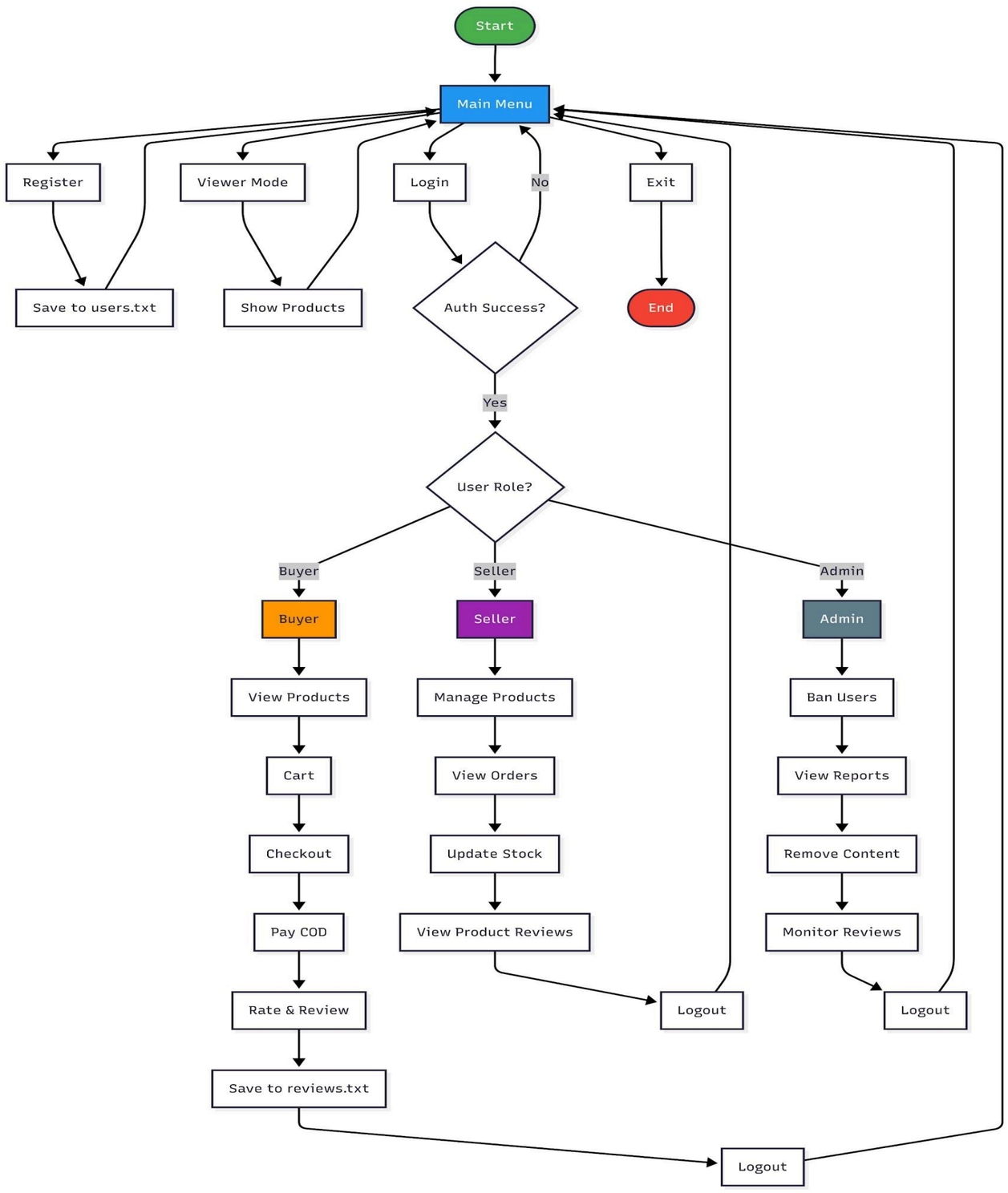
List of team members:
1. Tahmid Hafiz (230042136)
2. Arian Abedin (230042145)
3. Ibrahim Inyass (230042149)
4. Shahib Al-Kabir (230042157)

Overview of the Project:
The focus of this project is to act like a digital marketplace for three types of users: buyers, sellers and admins. Here, the buyers can view, filter, add products to cart and remove products from the cart, while the sellers can add, edit, and delete products and also check orders. Admins, on the other hand, may ban users and remove products from the list.

The Motivation behind the Project:
As demands were identified for simple, secure and reliable online shopping, and, as such, we aim to develop an accessible e-commerce platform for users in this project.

This project will have applications in many areas, some of which are:
- Local Store Management - Offline inventory and sales tracking for small shops
- Campus/Canteen Ordering - Food pre-ordering system for colleges/offices
- Library Book Issuing - Manage book borrowing, returns and member history
- Workshop/Event Registration - Sign-ups with payment and attendance tracking

Flow chart / Class Diagram:


Start
Main Menu
Register
Viewer Mode
Login
No
Exit
Save to users.txt
Show Products
Auth Success?
End
Yes
User Role?
Buyer
Seller
Admin
Buyer
Seller
Admin
View Products
Manage Products
Ban Users
Cart
View Orders
View Reports
Checkout
Update Stock
Remove Content
Pay COD
View Product Reviews
Monitor Reviews
Rate & Review
Logout
Logout
Save to reviews.txt
Logout

Key Features of the Project:

1. User Management:
    - Register
    - Login
    - Logout
    - Three roles: Buyer, Seller, Admin
    - Update profile

2. Product Management:
    - For Sellers:
        - Add products
        - Edit products
        - Delete products
    - For Buyers:
        - View Products
    - Product categories
    - Stock quantity

3. Cart Checkout:
    - Add to cart
    - Remove items
    - Update items
    - View cart
    - Checkout
    - Create order

4. Order Management:
    - Generate order ID
    - Order status (Pending/Delivered)
    - Buyer order history
    - Check orders for specific products (for Sellers)

5. Payment:

- Cash on Delivery (CoD)
- Payment confirmation

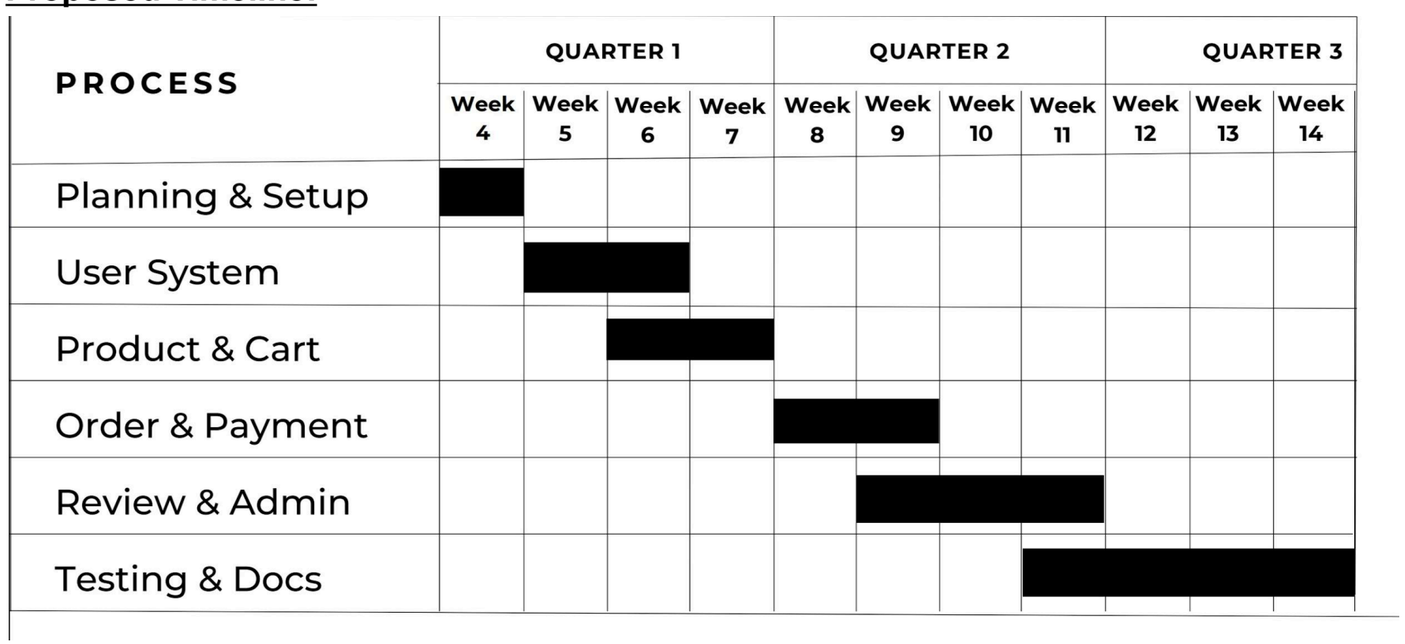
6. Review and Ratings:
    - Leave review
    - Leave rating

7. Admin Controls:
    - Approve Seller
    - Ban user
    - Remove product
    - View all orders

Tools and Technology:

- IDE: IntelliJ IDEA
- Github
- Programming Language: Java

Proposed Timeline:



Actions taken based on the feedback:

Based on the feedback received by our supervisor, Faisal Hussain, we have decided to implement the function of allowing the user to view their profile without logging in.